

Final Project Report

1. Introduction

1.1 Project Overview

Malnutrition among children under five remains a serious global health challenge that affects both physical and cognitive development. To better understand the scale and nature of this issue, this project analyzes global malnutrition trends from 1983 to 2019, focusing on key indicators such as stunting, wasting, underweight, and overweight.

Using **Power BI**, the analysis visualizes data from multiple countries categorized by income levels low, lower-middle, upper-middle, and high income as well as special classifications like LDC, LLDC, LIFD, and SIDS. The goal is to provide policymakers, NGOs, and researchers with an interactive visual tool that helps identify high-burden regions and supports targeted interventions.

1.2 Objectives

- Identify countries with the highest malnutrition burden.
- Compare malnutrition trends across income groups and global regions
- Provide actionable insights for policymakers, NGOs, and healthcare organizations.
- Develop an interactive Power BI dashboard that enables visual analysis of historical trends and patterns.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

Malnutrition continues to affect millions of children under the age of five globally, leading to long-term health, education, and economic consequences. Policymakers and organizations often struggle with fragmented and complex datasets that are difficult to interpret.

This project addresses that challenge by analyzing historical malnutrition data and building an interactive dashboard to pinpoint high-burden countries and understand correlations between income levels and malnutrition types.

2.2 Project Proposal (Proposed Solution)

- **Solution:** Develop a Power BI dashboard that visualizes malnutrition indicators across countries and income levels, enabling comparison, trend analysis, and data-driven decision-making.
- **Expected Outcomes:**
 - o A prioritized list of countries with high malnutrition burden.
 - o Visual insights into stunting, wasting, underweight, and overweight trends.
 - o An interactive tool for decision-makers to monitor progress over time.

2.3 Initial Project Planning

Task	Timeline	Responsibility	Status
Data Collection	1 October 2025	Self	Completed
Data Cleaning & Preprocessing	1 October 2025	Self	Completed
Dashboard Design & Visualization	2 October 2025	Self	Completed
Report Writing	2 October 2025	Self	Completed

Note: The entire project was successfully completed in two days (1–2 October 2025), including dataset processing, visualization, and documentation.

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources

- **Datasets Used:**
 1. country-wise averages.csv – Country-level malnutrition metrics.

2. malnutrition-estimates.csv – Detailed data on stunting, wasting, underweight, overweight for U5 children.

- Source: Kaggle Global Malnutrition Dataset.

3.2 Data Quality Report

- Checked for missing or inconsistent values.
- Corrected country names for proper mapping.
- Cleaned columns: Stunting_%, Wasting_%, Underweight_%, Overweight_%, Income_Group.
- Added calculated column $\text{Burden} = \text{U5 Population} \times \text{Stunting_} \% / 100$ for country-level impact.

3.3 Data Exploration and Preprocessing

- Computed averages for global and income group comparisons.
- Verified outliers and ensured consistent formatting for visualizations.
- Preprocessed data is Power BI ready for dashboards and analysis.

4. Data Visualization

4.1 Framing Business Questions

The dashboard was designed to answer key analytical questions such as:

- Which countries have the highest malnutrition burden?
- How do malnutrition indicators vary across income classifications?
- What are the long-term global trends in stunting, wasting, and underweight?
- Which regions require immediate intervention?

4.2 Developing Visualizations

Key visuals used in Power BI:

- **KPI Cards:** Displaying total survey sample size (11M), total underweight cases (2.08K), and average stunting.

- **Line Chart:** Showing trends of malnutrition indicators from 1983–2019.
- **Stacked Bar Chart:** Comparing stunting, wasting, and overweight rates across income classifications.
- **Ribbon Chart:** Displaying transitions of overweight vs. underweight by income group.
- **Map Visualization:** Highlighting country-wise distribution of stunting and underweight.
- **Treemap and Scatter Plots:** Showing the top 10 countries with highest stunting or wasting burden.

5. Dashboard

5.1 Dashboard Design File

The interactive Power BI dashboard consists of four main pages:

1. **Global Overview** – Displays global summary and key metrics.
2. **Country Analysis** – Allows drill-down into country-level data.
3. **Priority Countries** – Highlights top affected nations by indicator.
4. **Insights & Recommendations** – Summarizes findings and policy implications.

Features:

- Filters for Country, Year, and Income Group
- Drill-through options for detailed analysis
- Interactive slicers and tooltips for user engagement

6. Report

6.1 Story Design File

Each page in the dashboard is structured for storytelling:

- Key insights displayed using cards and visuals
- Comparisons across income groups and years
- Highlighted regions based on data trends
- KPIs summarizing main metrics (survey samples, stunting averages, etc.)

7. Performance Testing

7.1 Utilization of Data Filters

Applied interactive filters for Country, Year, and Income Group to dynamically analyze changes across dimensions.

7.2 Number of Visualizations

The final Power BI dashboard includes **12 key visuals**:

World Maps, Line Charts, Treemaps, KPI Cards, and Scatter Plots.

8. Conclusion / Observations

- Stunting and underweight prevalence remain high in low-income countries.
- India, Nigeria, and Pakistan contribute most to global stunting burden.
- Malnutrition is inversely related to income levels.
- Policy interventions needed for Sub-Saharan Africa and South Asia.
- Overweight prevalence rising in upper-middle-income countries, indicating double-burden malnutrition.

9. Future Scope

- Integrate additional datasets on maternal nutrition, vaccination coverage, and micronutrient deficiencies.
- Implement predictive modeling to forecast future malnutrition trends.
- Develop mobile-friendly dashboards for accessibility.
- Combine with broader global health and socioeconomic indicators for holistic insights.

10. Appendix

10.1 GitHub & Project Demo Link

- GitHub: [Link](#)
- Project Demo (Power BI file):  Project Demonstration.mp4